

Checksum Validation (CPP-002)

CPP-Identifier	CPP-002
CPP-Label	Checksum Validation
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Change history	Comments
Version 1.0.0 - 2025-08-29	Milestone version
Version 1.1.0 - 2025-10-16	Migration to XML

1. Description of the CPP

The TDA validates checksums against those stored in the Information Package at Ingest or Access.

Inputs and outputs

Input(s)	
Data	File
Metadata	Fixity metadata
Output(s)	
Metadata	Provenance metadata (the checksum validation event, including the datetime)
Alerts	File corruption alert for each File that fails the check
	Unsupported checksum algorithm alert for each algorithm that is not supported by the system

Definition and scope

The Checksum Validation process compares the checksums that come with the *Information Package*, with checksums based on the current state of the *File*'s data. Checksums of *Files* are validated at *Ingest* or *Access* to a TDA. The checksums may be either part of 1) a *SIP*; 2) an imported *AIP*; or 3) an *AIP* stored in the *TDA*.

A misalignment between the checksum algorithms used in the *Information Package* and the TDA's checksum policy's list of algorithms may occur. For example, incoming *Information Packages* (*SIPs* and *AIPs*) may include a checksum with a different algorithm created by the producer. In the case of *AIPs*, the checksum policy might have been updated during the lifecycle of the TDA, resulting in the removal and/or addition of algorithms. In such cases, the process should use whatever algorithm is associated with a given checksum to the best of its potential.

Process description

Trigger event(s)

Trigger Event	CPP-identifier
<i>SIP</i> or <i>AIP</i> import	CPP-029 (Ingest)
<i>DIP</i> generation	CPP-025 (Enabling Access)
<i>AIP</i> export	CPP-006 (AIP Batch Export)

Step-by-step description

No	Supplier	Input	Steps	Output	Customer
↓↑ sequence					
1	CPP-001 (Checksum Generation and Recording)	Fixity information	Get the list of pre-calculated checksums with their respective algorithms	List of checksums and algorithms	
		Fixity information from <i>SIP</i>			
2		List of checksums and algorithms	Evaluate for each algorithm individually if it is supported by the system.	Algorithm supported (step 3)	
		Digital archive system configuration		Algorithm not supported: - examine further procedure (e.g. based on legal agreements; submission policies; communication with producer, if possible) - Process completed	

No	Supplier	Input	Steps	Output	Customer
3		<i>File</i>	For each algorithm, recalculate the checksum of the <i>File</i> and match it with the given checksum	All <i>File</i> checksums match (step 5)	
		List of checksums and algorithms		Alert that any of the <i>File</i> checksums does not match: 1. Examine further procedure 2. Process completed	
5		All <i>File</i> checksums match	Document the event and its timestamp	Datetime for the checksum generation in the Provenance metadata	

Rationale(s) and worst case(s)

Rationale	Impact of inaction or failure of the process
Making sure the data content is still in the condition as intended	Data corruption during transfers into the system or during export could go unnoticed

2. Dependencies and relationships with other CPPs

Dependencies

CPP-ID	CPP-Title	Relationship description
CPP-001	Checksum Generation and Recording	CPP-002 relies on fixity information as produced and stored by CPP-001, but only during CPP-025 Enabling Access and CPP-006 AIP Batch Export. During CPP-029 Ingest, the fixity information supplied by the SIP will be used instead.

Other relations

Relation	CPP-ID	CPP-Title	Relationship description
Required by	CPP-006	AIP Batch Export	To ensure the integrity of the data during transport from the TDA storage the exported Files' checksums need to be verified.
Required by	CPP-025	Enabling Access	As the <i>DIP</i> is created, all <i>File</i> checksums need to be validated to ensure that the <i>DIP</i> is representative of the <i>AIP</i> .
Required by	CPP-029	Ingest	For any <i>SIP</i> submitted to the TDA all included <i>File</i> checksums need to be checked to validate the integrity of the <i>Files</i> .
Affinity with	CPP-004	Data Corruption Management	All new <i>AIP</i> copies must have their checksum validated to verify that the process was successful. However, the checksum validation is more mechanical in its nature, only aiming at verification of the copy process. The CPP-002 checksum validation is more comprehensive (including negotiations with producers and validation of results).
Affinity with	CPP-030	Refreshment	All new <i>AIP</i> copies must have their checksum validated to verify that the process was successful. However, the checksum validation is more mechanical in its nature, only aiming at verification of the copy process. The CPP-002 checksum validation is more comprehensive (including negotiations with producers and validation of results).
Affinity with	CPP-011	Replication	When <i>Files</i> are replicated, successful replication is validated by comparing the replicated <i>Files</i> ' checksums against the original <i>Files</i> ' checksums

Relation	CPP-ID	CPP-Title	Relationship description
Not to be confused with	CPP-003	Integrity Checking	Both CPPs get input from CPP-001, and both calculate a checksum from an <i>Information package</i> and compare it to a given checksum. The difference is that CPP-002 is done during the <i>Ingest</i> or <i>Access</i> phases (relating to transfer of content, changes in space), while CPP-003 is done periodically during the preservation of the contents in the archival storage (relating to changes over time). Thus, CPP-002 and CPP-003 are not only triggered by different processes, but also trigger different responses.

3. Links to frameworks

Certification

Certification framework	Term used in framework to refer to the CPP	Section
CTS	Checksum (cf Extended Guidance documentation)	Information Technology and Security/Storage & Integrity (R14)
Nestor Seal	/	/
ISO 16363	/	/

Other frameworks and reference documents

Reference Document	Term used in framework to refer to the process	Section
OAIS	Quality assurance	4.2.3.3 (Ingest)
PREMIS	/	/

4. Reference implementations

Publicly available documentation

Institution	Organisation type	Language	Hyperlink
TIB - Leibniz Information Centre for Science and Technology and University Library, DE	National library; Non-commercial digital preservation service; Research infrastructure; Research performing organisation	English	https://wiki.tib.eu/confluence/spaces/lza/pages/93608391/Preservation+of+data+integrity+as+part+of+the+process+routines
CSC - IT Center for Science Ltd., FI	Non-commercial digital preservation service	English	https://urn.fi/urn:nbn:fi-fe2024051731943 (Annex 3, section 2.1.1)
Archivematica, CA	Digital preservation system	English	https://www.archivematica.org/en/docs/archivematica-1.17/user-manual/transfer/transfer/#create-a-transfer-with-existing-checksums